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RF CMOS-MEMS CANTILEVER RESONATOR

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***Abstract*—This paper presents use of the simple cantilever beam as RF CMOS-MEMS resonator, which can be scaled like transistor with improved performance. This resonator can be used in the oscillator and filter part of RF transceiver, which can actually be fabricated on chip without losing its Q-factor. The cantilever beam resonator was designed using COMSOL4.3b™ and produce resonance frequency at 10.2MHz, pull-in voltage at 1.5V and Q of 10000.**

***Keywords*— resonators; oscillators; beam's displacement, gain,RF-MEMS**

I. INTRODUCTION

Micro-electro-mechanical systems (MEMS) have actively developed into research and development in a broader range of industrial application and academic pursuits. MEMS devices result from the integration of mechanical elements (resonators) and electronic components on a common silicon substrate using micro fabrication technology [3]. MEMS resonators have potential to replace quartz crystal and surface acoustic wave (SAW) resonator and are widely used to implement high-Q oscillators used in radio frequency (RF) industrial or communication transceivers [1]. At the magnitude higher quality factor Q, filters in RF of networking system that using MEMS technologies exhibits better performance than filters implemented using conventional transistor technologies in insertion loss, percent bandwidth, achievable rejection, and dynamic range[4]. MEMS resonators also provide on-chip components inherent with complementary metal–oxide–semiconductor (CMOS) process that must interface with transistor electronics at the board level and a solution to overcome the significant bottleneck against the ultimate miniaturization of wireless communication [4]. The utilize of these elements integrated into a single silicon chip, along with the CMOS circuitry, will improve the stability of the system performance with added benefits of reduced costs, power consumption and size [2]. This paper will explain and discuss designing of resonator and effects of various parameters on the resonator behavior.

II. BEAM STRUCTURE AND PRINCIPLE

The cantilever beam resonator has a single beam at the centre and the end of it is fixed to the anchors. Fig.1 shows the diagram of cantilever beam resonator 1. Fig.2 shows the resonator's structure and its electrical simulation setup. Aluminium (Al) material has been used to develop and design this resonators. The electrod, anchors and beam are designed using metal layers inherent in the CMOS technology process.

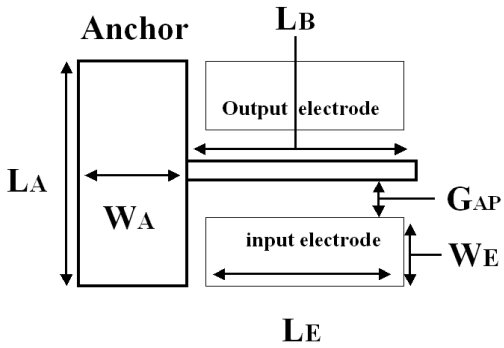

Fig. 1: Cantilever beam resonator.
The beam is fixed to end of the anchor.

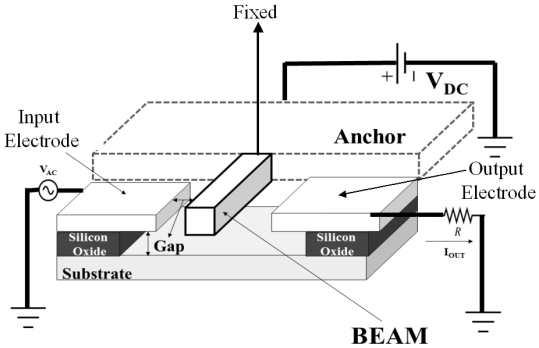

Fig. 2:MEMS filter using cantilever beam resonator

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The silicon layer acts as a ground plane.The silicon oxide is deposited underneath of anchors. Basically, the priciple of operation of a mechanical resonator is same as that of guitar string. A guitar string of 25'' made from Ni and steel tuned to a note. A vibrate at exactly 110Hz makes the guitar string is actually mechanically selecting this frequency, and is doing so with Q on the ordeer of 350, which is ~50 times more frequency selective than a typical on-chip electrical LC tank. Theb RF is also acting the same thing but at higher frequencies level. These frequencies can be achieved by scaling, similar to transistor technology. IC-compatible materials (such as titanium, polysilicon, and aluminium), supporting it nodes rather than its ends especially to minimize the anchor losses[7] . Consider a cantilever beam as shown in Fig. 2, the moving and fixed part forms two electrodes of capacitors. Table 1 shows all the design parameters of clamped-clamped resonator in this work. The electrical simulation setup of the resonator where the operation of the device is described when a direct current, DC and alternate current, AC voltage are applied to to the resonators. A DC actuation voltage is applied to the moving part of the anchor and AC signal is applied to the input electrode. The DC voltage is needed to create the necessary electrostatic force will bend but the beam will vibrate only if this force vibrates, which is done by applying an AC signal at the fixed electrode. The capacitance changes between the sense electrode and the beam due to the movement of the beam generates a corresponding current in the readout electrode[7]. The microstructure vibrating (motion) is detected by the sensing electrode, which produces an AC current at the sense electrode of the device. Hence, they are resulting in an electrostatic force between the electrode and the beam that consequently causes the vibration of the resonator. The capacitance change between the sense electrode and the beam due to the movement of the beam generates a current in the sense electrode.

III. MEMS Beam Resonators: Resonance Frequency

The resonance frequency (f_{res}) depends on the physical dimension of the device, shape, the vibration mode and the mechanical properties of the material [4]. The resonance frequency of the beam described in this work is 10MHz. The resonators are simulated using FEM have flexural and lateral resonance modes.

TABLE 1:
MEMS Resonator Design Parameters

Input Parameter	Value	Unit
Beam length, L_B	9	µm
Beam width, W_B	1	µm
Electrode width, W_E	1	µm
thickness, t	1	µm
Electrode to resonator Gap, GAP	0.5	µm
Anchor Length, L_A	22	µm
Anchor Width, W_A	10	µm

The beam oscillates with maximum amplitude at its resonance frequency:

$$\textbf{Cantilever}: f_{res} = \frac{1.03(W_B)}{(L_B)^2} \cdot \sqrt{\frac{E}{12\rho}} \quad (1)$$

Where E and ρ are the Young's modulus and density of the structural material, respectively. W_B is referred as width of beam resonator and L_B is the length of the resonator. According to the equation for resonance frequency the frequency increases as the length decreases. This oscillation of the beam changes between the electrodes also changing the capacitance between the two electrodes. The variation in capacitance produces an AC current by:

$$\textbf{AC current}: i = V_p \cdot \frac{dC}{dt} \quad (2)$$

The variation in capacitance is maximum at the resonance frequency. AC signal is applied to move electrode of first beam and the output current is measured at the output electrode.

III. Analysis using Comsol Multiphysics

The structure was done in 3D builder of the COMSOL™ Multhyphisic (version 4.3b). The material chosen for resonator is Aluminum (Al) having following material properties:

TABLE 2:
Material Properties of Aluminum

	Property	Name	Value	Unit	Property group
✓	Density	rho	2700[kg/...	kg/m^3	Basic
✓	Young's modulus	E	70.0e9[Pa]	Pa	Young's modulus and Poisson's ...
✓	Poisson's ratio	nu	0.35	1	Young's modulus and Poisson's ...
	Electrical conductivity	sigma	35.5e6[S...	S/m	Basic
	Coefficient of thermal expansion	alpha	23.1e-6[...	1/K	Basic
	Heat capacity at constant pressure	Cp	904[J/(kg...	J/(kg*K)	Basic
	Thermal conductivity	k	237[W/(...	W/(m*K)	Basic

The resonator was placed in an environment of air that having dimension 50µm x 50µm x 50µm. The material properties are:

TABLE 3:
Material Properties of Air

	Property	Name	Value	Unit	Property group
✓	Relative permittivity	epsilonr	1	1	Basic
	Relative permeability	mur	1	1	Basic
	Dynamic viscosity	mu	eta(T[1/K])[...	Pa*s	Basic
	Ratio of specific heats	gamma	1.4	1	Basic
	Electrical conductivity	sigma	0[S/m]	S/m	Basic
	Heat capacity at constant pressure	Cp	Cp(T[1/K])[J...	J/(kg*K)	Basic
	Density	rho	rho(pA[1/Pa]...	kg/m^3	Basic
	Thermal conductivity	k	k(T[1/K])[W/...	W/(m*K)	Basic
	Speed of sound	c	cs(T[1/K])[m/s]	m/s	Basic

A. Electrostatic Analysis

The electrostatic analysis was done to obtain spectrum of electrical parameters for resonator beam like static capacitance as function of voltage or pull in voltage. The physics included for analysis was Electromechanics (emi).

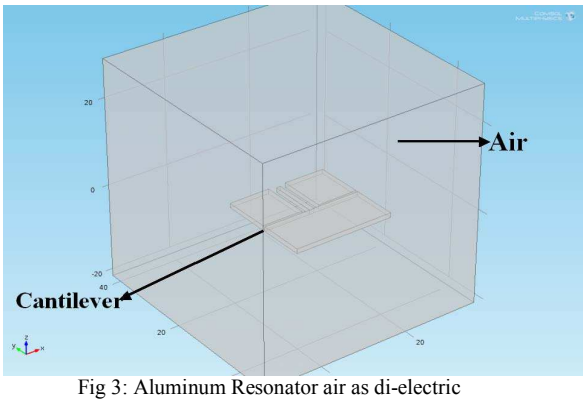

Fig 3: Aluminum Resonator air as di-electric

Electromechanics (emi), consists physical combination of Electrostatic (es), Solid mechanics (solid) and Moving Mesh (ale). The electrostatics part was applied to all the domain structure, the solid part to resonator beam, anchor and electrodes and finally the moving mesh (ale) part was applied to the beam only. Then DC signal is applied as a parameter to terminal of moving electrode and ground is selected at the bottom of electrodes As DC signal is applied to the anchor , the electric field is generated which puts a force on the moving part. The moving mesh evaluates the displacement of the beam as a function of applied potential. In the study the parameter DC signal was varied from 0V to 6V. The equations defining the issue are standard Maxwell equation and Newton's equation [6]:

$$\nabla.D = \rho.V \quad (3)$$

$$E = -\nabla.V \quad (4)$$

When solved in COMSOL, the following result is obtained:

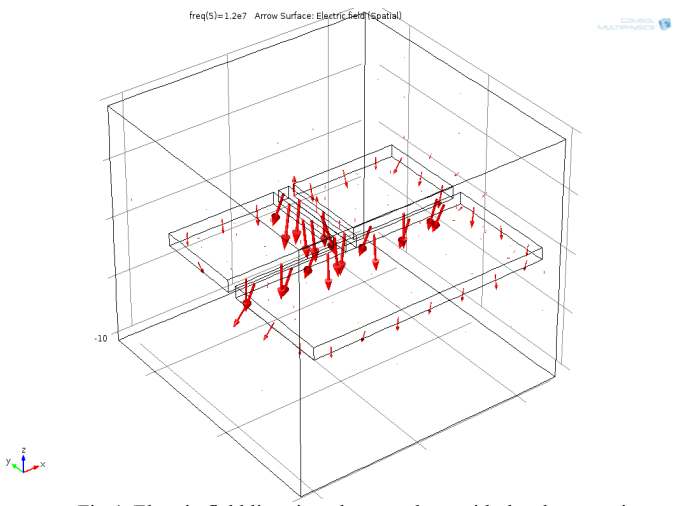

Fig.4: Electric field lines in red arrow along with the electrostatic displacement of the beam.

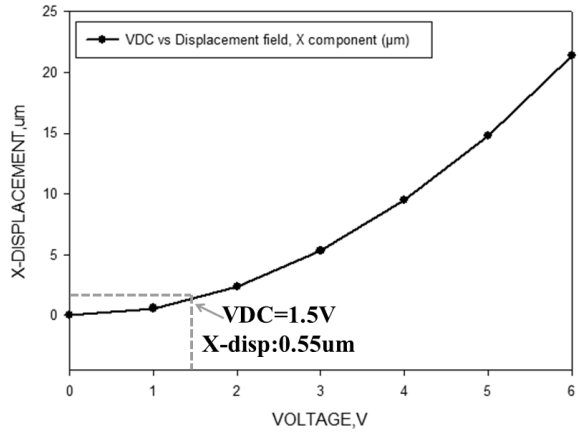

Fig.5: Displacement as function of DC voltage

The pull-in voltage is given by [5] :

$$Vpull-in = \sqrt{\frac{8kd^3}{27\varepsilon A}}$$

Analytically, it exhibits about 1.3V. The pull-in voltage obtained from Fig.5 is 1.5V.

B. Eigeinfrequency Analysis

The stationary waves on a cantilever structure are quantized meaning the frequencies corresponding to form stationary waves are quantized. The lowest frequency for which these stationary waves are formed is called fundamental harmonic. The modes of oscillation have different shapes for different frequencies. These eigen frequencies are obtained by solving the eigen value equation of the form[5]:

$$AV = \lambda BV \quad (5)$$

The eigen value results and corresponding modes of the beam oscillation are shown in simulation results for first three modes.

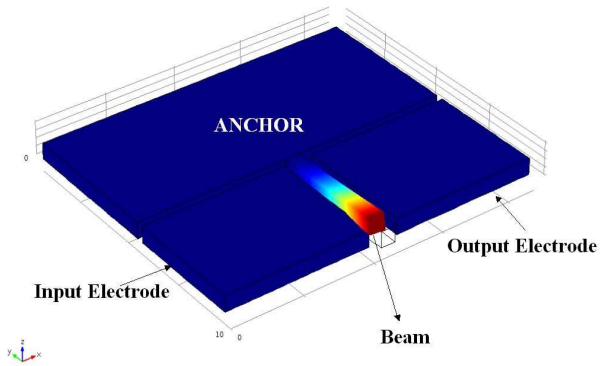

Fig. 6: The cantilever at X,Y,Z dimension

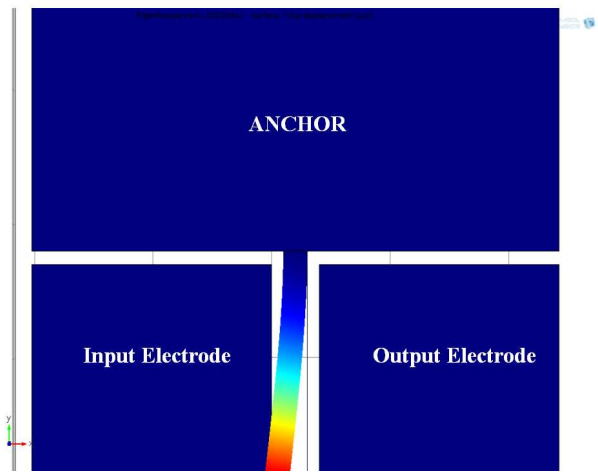

Fig. 7: The cantilever at X, Y condition. The beam moves laterally

The calculated value of the resonance frequency is 9.69 MHz but the simulated frequency is 10.23 MHz for mode 1. This is because some of the energy is lost in damping at the anchors [6].

C. Frequency Domain Analysis

The frequency's resonator that works at very high Q oscillator is obtained by applying and AC signal at one of the electrode. The frequency range is varied of this AC source. The resonance curve is observed to be a function of the static voltage. As the static voltage increases the internal stress of the beam also increase. This is because of the increment to an increase in electrical stiffness and decrease in resonance frequency. The resonance curve shifts, as the DC signal is change. The resonance curve of Fig.8 is obtained for DC voltage of 1V and provided the frequency at 10.2MHz. In term of thermo-Elastic damping at the anchors the environment when the beam oscillates the amount of displacement is maximum at the center of the beam and decreases as one traverse towards the anchors. Thus the energy conservation from potential to kinetic is followed at other nodes except at the anchors. At the anchors the energy is transferred from anchors to substrate in the form of heat. This causes anchors to heat and also decreases the Q of the beam. The mechanism of thermo-elastic damping is one of the reasons why the power handling capacity of the beam is less than what is expected.

The Q factor of this resonator can be estimated as:

$$Q = \frac{fres}{BW} \qquad (6)$$

This comes out about 10,000 in vacuum, which is very high as compared to a standard filter fabricated using transistors.

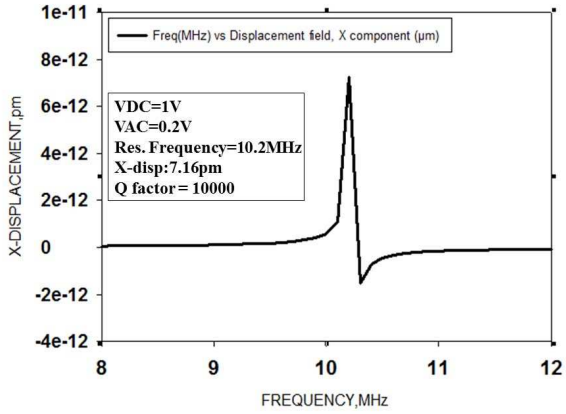

Fig. 8 : Maximum displacement as function of frequency

IV. Conclusion

The cantilever resonator beam is designed and presented. It was found to have Q of 10,000 and resonance of 10.23MHz, which have high potential to have higher values. The cantilever resonator also was built using Aluminum (Al), which can be easily fabricated by the standard CMOS process and can be used in on chip integrated circuit. Finally the pull-in voltage of this resonator is 1.5V and this is CMOS-compatible.

Acknowledgment

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